

Arrays

21 Interview Questions — Easy to Hard

The Complete DSA + SQL Interview Preparation Series

Book 1 of 13

For placement preparation, coding interviews, and product-based company rounds.

About This Book

This book covers arrays, the single most common category of coding interview questions. Each question here is kept short and direct: just the problem, and a sample input/output pair written in plain array notation so you can see exactly what's expected — no constraints tables, no hints, no scaffolding.

Read each problem the way it would sound in a live interview and start reasoning immediately.

How To Use This Book

Read each problem once. Restate it in your own words before writing any code. Use the sample input/output only to confirm your understanding of the expected behavior — not as a spec to reverse-engineer from. Try to solve every problem in your head or on paper before typing anything.

Difficulty Breakdown

Difficulty	Count
Easy	5
Medium	11
Hard	5

Table of Contents

Q1.	Two Numbers That Add Up	Easy
Q2.	Find the Missing Number	Easy
Q3.	Reverse the Array	Easy
Q4.	Best Time to Buy and Sell	Easy
Q5.	Detect a Duplicate	Easy
Q6.	Range Sum Query	Medium
Q7.	Maximum Sum Subarray of Size K	Medium
Q8.	Product of Array Except Self	Medium
Q9.	Rotate the Array by K	Medium
Q10.	Merge Overlapping Intervals	Medium
Q11.	Sort an Array of 0s, 1s, and 2s	Medium
Q12.	Find the Majority Element	Medium
Q13.	Maximum Subarray Sum	Medium
Q14.	Search in a Rotated Sorted Array	Medium
Q15.	Split Array Into Chunks of Size K	Medium
Q16.	Trapping Rain Water	Hard
Q17.	Maximal Rectangle in a Binary Matrix	Hard
Q18.	Sliding Window Maximum	Hard
Q19.	Minimum Platforms Needed	Hard
Q20.	First Missing Positive Number	Hard
Q21.	Median of Two Sorted Arrays	Hard

Q1. Two Numbers That Add Up

Easy

Topic: Two Sum

Given an array of prices and a target, find the indices of the two items whose prices add up exactly to the target.

INPUT

```
prices = [2,7,11,15], target = 9
```

OUTPUT

```
[0, 1]
```

Q2. Find the Missing Number

Easy

Topic: Missing Number

An array holds N-1 distinct numbers drawn from the range 1 to N. Find the one missing number.

INPUT

```
arr = [1,2,4,5], N = 5
```

OUTPUT

```
3
```

Q3. Reverse the Array

Easy

Topic: Two Pointers

Reverse an array in place, without allocating a second array.

INPUT

```
arr = [1,2,3,4,5]
```

OUTPUT

```
[5,4,3,2,1]
```

Q4. Best Time to Buy and Sell

Easy

Topic: Sliding Window

Given daily stock prices, find the maximum profit from buying on one day and selling on a later day. Return 0 if no profit is possible.

INPUT

```
prices = [7,1,5,3,6,4]
```

OUTPUT

```
5
```

Q5. Detect a Duplicate

Easy

Topic: Duplicate Detection

Given an array, determine whether any value appears more than once.

INPUT

```
arr = [1, 2, 3, 4, 2]
```

OUTPUT

```
true
```

Q6. Range Sum Query

Medium

Topic: Prefix Sum

Given an array and multiple (L, R) range queries, return the sum of elements from index L to R (inclusive) for each query, answered in O(1) after preprocessing.

INPUT

```
arr = [1, 2, 3, 4, 5], queries =  
[(0, 2), (1, 4)]
```

OUTPUT

```
[6, 14]
```

Q7. Maximum Sum Subarray of Size K

Medium

Topic: Sliding Window

Given an array and an integer k, find the maximum sum of any k consecutive elements.

INPUT

```
arr = [2, 1, 5, 1, 3, 2, 7], k = 3
```

OUTPUT

```
12
```

Q8. Product of Array Except Self

Medium

Topic: Product Except Self

Given an array, return a new array where each element is the product of all the other elements, without using division.

INPUT

```
arr = [1, 2, 3, 4]
```

OUTPUT

```
[24, 12, 8, 6]
```

Q9. Rotate the Array by K

Medium

Topic: Rotate Array

Rotate an array to the right by k positions, in place, without allocating a second array.

INPUT

```
arr = [1, 2, 3, 4, 5, 6, 7], k = 3
```

OUTPUT

```
[5, 6, 7, 1, 2, 3, 4]
```

Q10. Merge Overlapping Intervals

Medium

Topic: Merge Intervals

Given a list of intervals, merge all intervals that overlap or touch, and return the merged, non-overlapping intervals.

INPUT

```
intervals =  
[[1, 3], [2, 6], [8, 10], [15, 18]]
```

OUTPUT

```
[[1, 6], [8, 10], [15, 18]]
```

Q11. Sort an Array of 0s, 1s, and 2s

Medium

Topic: Dutch National Flag

Given an array containing only 0s, 1s, and 2s, sort it in place in a single pass, without using a general-purpose sort.

INPUT

```
arr = [2, 0, 2, 1, 1, 0]
```

OUTPUT

```
[0, 0, 1, 1, 2, 2]
```

Q12. Find the Majority Element

Medium

Topic: Majority Element

Given an array where one value appears more than $N/2$ times, find that value using constant extra space.

INPUT

```
arr = [2, 2, 1, 1, 1, 2, 2]
```

OUTPUT

```
2
```

Q13. Maximum Subarray Sum

Medium

Topic: Kadane's Algorithm

Find the maximum possible sum of any contiguous subarray.

INPUT

```
arr = [-2, 1, -3, 4, -1, 2, 1, -5, 4]
```

OUTPUT

```
6 ([4, -1, 2, 1])
```

Q14. Search in a Rotated Sorted Array

Medium

Topic: Binary Search

A sorted array has been rotated an unknown number of times. Given a target, find its index in $O(\log N)$, or return -1 if not found.

INPUT

```
arr = [4, 5, 6, 7, 0, 1, 2], target = 0
```

OUTPUT

```
4
```

Q15. Split Array Into Chunks of Size K

Medium

Topic: Array Chunking

Given an array and an integer k, split it into consecutive groups of size k, with the final group holding whatever elements remain.

INPUT

```
arr = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10], k = 3
```

OUTPUT

```
[[1, 2, 3], [4, 5, 6], [7, 8, 9], [10]]
```

Q16. Trapping Rain Water

Hard

Topic: Trapping Rain Water

Given an array representing bar heights, compute how much water would be trapped between the bars after it rains.

INPUT

```
height = [0, 1, 0, 2, 1, 0, 1, 3, 2, 1, 2, 1]
```

OUTPUT

```
6
```

Q17. Maximal Rectangle in a Binary Matrix

Hard

Topic: Matrix Problems

Given a binary matrix, find the area of the largest rectangle containing only 1s.

INPUT

```
matrix = [[1,0,1,0,0],[1,0,1,1,1],[1,1,1,1,1],[1,0,0,1,0]]
```

OUTPUT

6

Q18. Sliding Window Maximum

Hard

Topic: Monotonic Stack

Given an array and window size k, return the maximum value in every window of size k as it slides from left to right.

INPUT

```
arr = [1,3,-1,-3,5,3,6,7], k = 3
```

OUTPUT

```
[3,3,5,5,6,7]
```

Q19. Minimum Platforms Needed

Hard

Topic: Greedy with Arrays

Given train arrival and departure times, find the minimum number of platforms needed so no two overlapping trains ever share one.

INPUT

```
arrival =  
[900,940,950,1100,1500,1800],  
departure =  
[910,1200,1120,1130,1900,2000]
```

OUTPUT

3

Q20. First Missing Positive Number

Hard

Topic: Advanced Array Manipulation

Given an unsorted array that may contain negatives, zeros, and duplicates, find the smallest missing positive integer in $O(N)$ time and $O(1)$ extra space.

INPUT

`arr = [3, 4, -1, 1, -8]`

OUTPUT

2

Q21. Median of Two Sorted Arrays

Hard

Topic: Binary Search

Given two sorted arrays, find the median of the combined dataset without fully merging them.

INPUT

`a = [1, 3], b = [2, 4, 5]`

OUTPUT

3